AUDITED COMPANION EDITION

Lambda as Complementary Modeling

Registry-Relative Symmetry, Translation Residue, and the Baryon-Photon Ratio

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Independent laboratory notes and research record

STATUS Publication candidate after internal editorial, logical, source, layout, and Lean-kernel audit. This is not peer review, not a replacement cosmology, and not a certification of Lambda/NSAF/TUFT physics.

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Companion to Before the Proof, Sections 12 and 19

Abstract

This companion holds one registered empirical estimate fixed while auditing the additional commitments by which that estimate becomes an explanatory problem. The case is the baryon-to-photon ratio. The paper distinguishes a conventional physical mechanism-seeking problem from a Lambda-side registry-audit problem; the latter is not presented as an alternative baryogenesis mechanism. A finite translation table now identifies the exact shared items, the source-side omissions, and the target-side methodological residue. The central non-uniqueness claim is stated conditionally and proved from explicit counter-witnesses rather than written as an inequality between unlike semantic objects. A Lean 4 appendix checks the abstract partial-map, residue, invariant, compatibility, and non-uniqueness relations. Empirical adequacy, physical interpretation, causal attribution, ontology, and the formalization seam remain outside Lean's jurisdiction.

> VERIFICATION BOUNDARY Lean checks the distributed formal source relative to its definitions and premises. It does not verify Big-Bang nucleosynthesis, the quoted estimate, baryogenesis, the Sakharov conditions, TUFT, or the claim that the formal source exhausts the prose.

Revision and provenance record

Artifact	Role	SHA-256
Unaudited companion PDF	Preserved source; not overwritten	EA9810E9E14BF39EB7443C6CBCD78F78E127AC4550429EE7B1608 2F3A980BC03
Parent paper R2 PDF	Frozen pairing authority	224888DA1FD46A7F53C13EC006035BF9E3B6FA617BD244BC1AD7 9FAE9A15D20A
LambdaCompanion_Audit.lean	Distributed executable claim kernel	2E9A8955F424E6CA5B512524231672908B0408E0C69830B82BCC8 904775FD3AE

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1. Scope and relation to Before the Proof

Before the Proof defines a registration chain $X \to M \to D \to B_{in} \to P \to F \to Q \to B_{out} \to I$ and separates verification of a declared traversal from verification of the bridges surrounding it. Its Sections 12 and 19 reserve the baryon-to-photon ratio as a companion case in which a registered estimate remains fixed while problem-forming commitments are exposed. This edition performs that task as a typed dependency audit.

The paper makes no new measurement, proposes no replacement cosmology, and offers no microphysical account of baryon production. It does not infer that a conventional explanation is false merely because an alternative audit question can be formed. It asks where each premise enters, what a declared translation preserves, and what remains outside that translation.

The governing discipline is non-promotion and non-demotion. Model-introduced structure is not promoted to a uniquely necessary physical cause merely because it supports a successful traversal. It is also not demoted to fiction merely because another registry does not use it as a primitive.

2. Fixed empirical anchor and registration seam

Let n_b and $n_{\bar{b}}$ denote baryon and antibaryon number densities. Define the net baryon density and baryon-to-photon ratio by

$$n_B := n_b - n_{\bar{b}}, \quad \eta_B := n_B / n_\gamma.$$

After baryon-antibaryon annihilation, $n_{\bar{b}}$ is much smaller than n_b, so η_B is approximately n_b/n_γ. This approximation is not the definition, and it should not be conflated with electron-positron annihilation or with the thermal bookkeeping required to compare photon densities across epochs.

The numerical anchor inherited from the parent paper is the 2024 Particle Data Group table entry labeled baryon-to-photon ratio (from BBN):

$$\hat{\eta}_B = (6.04 \pm 0.12) \times 10^{-10}. \; [2, 3]$$

The hat is material. The quantity is a BBN-inferred, model-mediated estimate obtained from abundance observations through likelihood, nuclear and thermal modeling, calibration, conventions, and parameter extraction. The value is frozen here for comparability with the parent paper; it is not asserted to be an unmediated observation or a uniquely privileged current estimate.

$$D\_\eta := (\text{value, uncertainty, provenance, registration contract}).$$

> EMPIRICAL JURISDICTION This paper registers D_η and cites its source. It does not reproduce the BBN likelihood analysis or certify the empirical truth of that analysis.

3. Two explicitly different problem types

3.1 Conventional mechanism-seeking problem

A conventional baryogenesis problem does not follow from D_η alone. It combines the registered estimate with a physical registry containing a cosmological history, dynamical ordering, conservation structure, symmetry assumptions, correspondence rules, and a mechanism-seeking explanatory target. Write

$$P\_std := Form\_std(D\_\eta; H\_hot, O\_dyn, S\_phys, C\_phys).$$

Within the conventional class of baryogenesis mechanisms for which the assumptions apply, the Sakharov conditions organize the search around baryon-number violation, C and CP violation, and departure from thermal equilibrium [4]. This paper neither re-derives nor challenges that program.

3.2 Lambda-side registry-audit problem

The Lambda-side problem has a different output type. It does not ask for a competing particle-production mechanism. It asks for a dependency record: which elements of P_std are contained in the shared registration, which are additional problem-forming commitments, which cross a declared translation, and which remain as directional residue. Write

$$P\_\Lambda := Audit\_\Lambda(D\_\eta; \text{source registry, target registry, partial translation, jurisdiction}).$$

A valid output of P_Λ is therefore an auditable classification, not a second cosmological prediction. Treating these outputs as interchangeable would reproduce the type collapse the parent method is designed to prevent.

3.3 Status of Lambda, NSAF, and TUFT terminology

Only the Lambda-side audit discipline is operational in this worked case. NSAF and TUFT-labeled materials are not premises of the formal kernel and are not used to derive η-hat_B or a baryogenesis mechanism. The original draft's phrase single Lambda/NSAF/TUFT registry is therefore narrowed: G_Λ denotes the author-side audit registry defined in this paper.

Jennifer Lorraine Nielsen's independently authored TUFT manuscript is cited solely to preserve provenance for the independently used TUFT name and Hopf-fibration program [7]. This paper does not audit, adopt, certify, reject, or attribute its Lambda traversal to Nielsen. No journal or peer-review status is asserted here.

4. Typed registries, partial translation, and residue

A registry is not treated as a bare set. For the present audit, let each registry carry a declared carrier, an admissibility predicate, a problem constructor, and an output projection:

$$G\_i := (S\_i, Adm\_i, Form\_i, q\_i).$$

The translation acts on carrier elements, not on the registry tuple itself:

$$\tau\_std\text{->}\Lambda : S\_std \text{ -> } Option(S\_\Lambda).$$

Writing Option makes partiality explicit. If τ(x) is undefined, x is outside the declared domain. No falsity, inconsistency, or physical unreality follows from that scope decision.

$$SrcRes(\tau) := \{ x \text{ in } S\_std \mid \tau(x) \text{ is undefined } \}.$$

$$TgtRes(\tau) := \{ y \text{ in } S\_\Lambda \mid \text{for every } x \text{ in } S\_std, \tau(x) \text{ is not } y \}.$$

These are directional and answer different questions. Source residue records source commitments omitted by the translation. Target residue records target structure that was not supplied as an image of any source item.

$$Preserves\_q(\tau, x) := \text{if } \tau(x) = y, \text{ then } q\_\Lambda(y) = q\_std(x).$$

5. Worked finite translation

The following finite declaration is the concrete worked case. It is intentionally small enough to inspect and exactly matches the constructors in the distributed Lean source.

Source item in S_std	τ_std->Λ	Target item in S_Λ	Audit meaning
Registered BBN estimate	mapped	Registered estimate	Shared empirical anchor; anchor projection preserved.
Exact ratio definition	mapped	Exact ratio definition	Same algebraic definition on both sides.
Hot Big-Bang history	undefined	-	Source-side physical history is outside this map's declared domain.
Baryon-number violation	undefined	-	Conventional mechanism condition; not translated or rejected.
C and CP violation	undefined	-	Conventional mechanism condition; not translated or rejected.
Departure from equilibrium	undefined	-	Conventional mechanism condition; not translated or rejected.

$$\mathrm{SrcRes}(\tau) = \{\text{hot history, B violation, C/CP violation, nonequilibrium}\}.$$

On the target side, datum/problem separation, the jurisdiction record, and the residue record are not images of source items in this finite map. They therefore form the declared target-side methodological residue.

$$\mathrm{TgtRes}(\tau) = \{\text{datum/problem separation, jurisdiction record, residue record}\}.$$

The registered estimate supplies the invariant witness: the source registeredEstimate item maps to the target registeredEstimate item, and both anchor projections return the same D_η record. The Lean theorem registered_anchor_invariant checks this exact surrogate.

6. Conditional non-uniqueness results

The original slogan is replaced by two typed propositions. Let Generates(g,p) mean that registry g generates problem p under a declared formation contract, and let Solved(p) mean that p has been solved under its own adjudication contract. Define

$$\mathrm{UniqueGenerator}(\mathrm{Generates}, g, p) := \text{for every } g', \mathrm{Generates}(g',p) \text{ implies } g' = g.$$

$$\mathrm{UniqueProblemFrom}(\mathrm{Generates}, d, p) := \text{for every } g',p', \mathrm{Generates}(g',d,p') \text{ implies } p' = p.$$

Proposition 6.1 - Generator non-uniqueness from a witness

If Solved(p), Generates(g,p), Generates(g',p), and g' is not g, then p is solved and g is not the unique generator of p. Proof: the supplied g' contradicts the universal condition in UniqueGenerator.

Proposition 6.2 - Problem non-uniqueness from a witness

If Solved(p), Generates(g,d,p), Generates(g',d,p'), and p' is not p, then p is solved and p is not the unique generated problem from d. Proof: the supplied pair (g',p') contradicts the universal condition in UniqueProblemFrom.

> SCOPE OF THE RESULT Neither proposition manufactures its counter-witness. Any real application must separately justify the Generates premises and the distinctness premise. Solving a problem alone is not enough; the non-uniqueness conclusion follows only when the additional witness is supplied.

The corresponding Lean theorems are solved_does_not_certify_unique_generator and solved_does_not_certify_unique_problem. Their names say exactly what the formal proof does and avoid the earlier category mismatch between a solution and a proof.

7. Joint closure as compatibility of projections

The standard traversal and the registry-audit traversal return different full output types. Joint closure is therefore defined only after declaring comparable projections. Let q_std(Q_std) and q_Λ(Q_Λ) be their registered-anchor projections. Then

$$\mathrm{Compat\_D}(Q\_std,Q\_\Lambda) := q\_std(Q\_std) = D\_\eta \text{ and } q\_\Lambda(Q\_\Lambda) = D\_\eta.$$

The witness consists of the two displayed equalities. It establishes agreement at the declared anchor projection only. It does not establish identity of primitives, equivalence of full outputs, physical adequacy of G_Λ, or reduction of one ontology to another. The Lean theorem joint_closure_witness checks the even narrower abstract form in which both scoped output projections equal one anchor.

8. Symmetry, causal language, and bridge discipline

An asymmetry is relative to a declared comparison structure: an equivalence relation, transformation class, or invariant specifies what would count as symmetry. Physical symmetries are not thereby arbitrary. Their roles can be experimentally supported regularities, exact formal invariances, effective approximations, boundary conditions, or explanatory ideals. Those roles should not be collapsed.

A mechanism derived under a symmetry-bearing physical registry can establish consequences of the stated premises. The derivation does not, by notation alone, show that the registered numerical estimate independently contains those premises or that the chosen causal decomposition is unique. Conversely, locating a premise in the problem-forming bridge does not refute or fictionalize it.

9. Control example: radix and tuning

Let a/b be a reduced rational number with b > 0 and let r >= 2 be an integer base. Its positional expansion terminates in base r exactly when b divides r^k for some nonnegative integer k. Equivalently, every prime factor of b is a prime factor of r. This follows because a terminating k-digit expansion has denominator r^k, while b divides r^k after reduction.

Thus 1/3 terminates in base 12 as 0.4(base 12) but repeats in base 10; 1/5 terminates in base 10 but repeats in base 12. The represented rational number does not change. The registry changes representational exactness.

Twelve-tone equal temperament divides the octave into twelve equal logarithmic steps, so the semitone frequency ratio is 2^(1/12). Pythagorean tuning instead builds intervals from powers of the 3/2 fifth [8]. Writing equal-tempered ratios in base 12 does not convert their generally irrational values into integer harmonic ratios. Physical division, tuning rule, and numeral representation remain differently typed objects.

10. Claim ledger

Claim or object	Type	Status	Adjudication boundary
η_B := n_B/n_γ	Formal definition	Declared exactly	Definition only; physical interpretation is separate.
η-hat_B = (6.04 +/- 0.12)x10^-10	Empirical inference	Externally sourced	PDG BBN table entry; not re-derived.
Sakharov conditions	Physical/theoretical	External background	Not challenged or machine-checked here.
P_std	Physical problem formation	Declared schema	Depends on physical history and mechanism premises.
P_Λ	Methodological audit	Operational here	Not a competing baryogenesis mechanism.
τ_std->Λ	Formal surrogate	Lean-checked	Only the declared finite carrier map is checked.
SrcRes and TgtRes	Formal definitions	Lean-checked witnesses	No truth or ontology verdict follows.
Anchor invariant	Formal equality	Lean-checked	Equality of encoded anchor projections only.
Non-uniqueness results	Conditional logic	Lean-checked	Require explicit distinct counter-witnesses.
Joint closure	Compatibility relation	Lean-checked surrogate	Agreement at anchor projection only.
Causal reading of symmetry	Bridge claim	Outside Lean	Requires separate physical warrant.
Nielsen TUFT	Independent external work	Cited, not audited	No endorsement, collaboration, or transfer of claims.

11. Lean-checked receipt

> STATUS: LEAN-CHECKED - DECLARED FORMAL JURISDICTION ONLY Checked: the distributed source elaborates with zero error markers and zero warning markers, and the encoded conclusions follow from the encoded premises. Not checked: empirical truth, semantic fidelity beyond the declared seam, physical interpretation, causation, ontology, model completeness, or external source claims.

Receipt field	Recorded value
Public service	Lean Web, https://live.lean-lang.org/
Project	mathlib-stable
Lean version	4.33.0, reported by #eval Lean.versionString
Imports	None; Lean core declarations only
Source file	LambdaCompanion_Audit.lean
Source SHA-256	2E9A8955F424E6CA5B512524231672908B0408E0C69830B82BCC8904775FD3AE
Result	0 errors; 0 warnings; 7 informational messages
Evidence capture	2026-08-22T05:49:45Z (UTC)
Persistent run	Compressed source URL recorded in the external manifest and evidence JSON

Verification evidence is distributed as LEAN_WEB_VERIFICATION_EVIDENCE.json. It records the exact source digest, public-service URL, stable project, Lean version, run interval, persistent source URL, and the seven informational outputs with zero error and zero warning diagnostics. Reproduction requires opening the distributed source in the named public service and confirming the same result. The public service processes submitted code; readers who prefer not to transmit source may use a local Lean 4.33.0 installation.

12. Conclusions and exclusions

The baryon-to-photon ratio provides a compact registry audit because the registered estimate can remain fixed while the formation of explanatory problems is made explicit. The worked result is now concrete: two shared carrier items are translated, four source physical commitments remain outside the map's domain, three target methodological objects remain outside its image, and the anchor projection is preserved.

Conventional baryogenesis remains a legitimate and highly constrained physical research program. This companion neither derives nor replaces it. The Lambda-side traversal is a dependency and jurisdiction audit. It acquires additional physical force only if a future work supplies a physical model, a separately audited translation, and discriminating empirical consequences.

The strongest supported conclusion is conditional and relational: solving one well-formed problem does not by itself certify uniqueness of its generator or uniqueness of the admissible problem formed from a datum. Those non-uniqueness conclusions require explicit counter-witnesses, and the Lean kernel checks precisely that limited logical structure.

> NOT ESTABLISHED No new η_B value; no Lambda/NSAF/TUFT baryogenesis mechanism; no rejection of Sakharov's conditions; no claim that symmetry is merely conventional; no new physical cause; no empirical equivalence; no certification of Nielsen's TUFT; and no claim that Lean verified the whole paper.

Appendix A - Exact distributed Lean source

The following source is reproduced verbatim from LambdaCompanion_Audit.lean, SHA-256 2E9A8955F424E6CA5B512524231672908B0408E0C69830B82BCC8904775FD3AE. Line wrapping in this appendix is typographic only. The external .lean file is the executable authority.

```
/-!
Lambda as Complementary Modeling - executable claim kernel

Scope: this file formalizes only the abstract registry-audit relations used in
the companion paper: a partial translation, directional source and target
residue, preservation of a registered anchor on the translated domain, and two
non-uniqueness consequences from explicit counter-witnesses.

It does not formalize or verify Big-Bang nucleosynthesis, the numerical
adequacy of the quoted estimate, baryogenesis, the Sakharov conditions,
Lambda/NSAF/TUFT physics, causal or ontological interpretations, or the
fidelity of the natural-language-to-Lean formalization seam.
-/

namespace LambdaCompanion

/-- A decimal scientific-notation record.  The intended display uses the
    central value, its stated uncertainty, and a power of ten. -/
structure EmpiricalAnchor where
  central : Nat
  uncertainty : Nat
  exponent : Int
deriving DecidableEq, Repr

/-- The paper's frozen 2024 PDG table entry, represented by central value 604,
    uncertainty 12, and exponent negative 12 (equivalently, central value
    6.04, uncertainty 0.12, and exponent negative 10).
    The source and empirical adequacy of this record are outside this file. -/
def bbn2024 : EmpiricalAnchor :=
  { central := 604, uncertainty := 12, exponent := -12 }

/-- Declared source-side items in the worked audit. -/
inductive StdItem where
  | registeredEstimate
  | ratioDefinition
  | hotBigBangHistory
  | baryonNumberViolation
  | cAndCpViolation
  | nonequilibrium
deriving DecidableEq, Repr

/-- Declared target-side items in the worked audit. -/
inductive LambdaItem where
  | registeredEstimate
  | ratioDefinition
  | datumProblemSeparation
  | jurisdictionRecord
  | residueRecord
deriving DecidableEq, Repr

/-- The deliberately partial translation used by the paper's finite worked
    example.  Only the shared datum and exact ratio definition are translated.
    `none` is a scope statement, not a verdict that the source item is false. -/
def translate : StdItem -> Option LambdaItem
  | .registeredEstimate => some .registeredEstimate
  | .ratioDefinition => some .ratioDefinition
  | .hotBigBangHistory => none
  | .baryonNumberViolation => none
  | .cAndCpViolation => none
  | .nonequilibrium => none

/-- Source-side omission: a declared source item is outside the map's domain. -/
def SourceResidue {A : Type} {B : Type} (tau : A -> Option B) (a : A) : Prop :=
  tau a = none

/-- Target-side residue: a declared target item is not hit by the map. -/
def TargetResidue {A : Type} {B : Type} (tau : A -> Option B) (b : B) : Prop :=
  forall a, Not (tau a = some b)
```

```

/-- Observable projection on the source carrier.  Only the registered estimate
    carries the empirical anchor in this small formal surrogate. -/
def stdAnchor : StdItem -> Option EmpiricalAnchor
  | .registeredEstimate => some bbn2024
  | _ => none

/-- Observable projection on the target carrier. -/
def lambdaAnchor : LambdaItem -> Option EmpiricalAnchor
  | .registeredEstimate => some bbn2024
  | _ => none

/-- The registered estimate is invariant on the translated point. -/
theorem registered_anchor_invariant :
    translate .registeredEstimate = some .registeredEstimate /\
      lambdaAnchor .registeredEstimate = stdAnchor .registeredEstimate := by
  exact And.intro rfl rfl

/-- A representative source-side residue witness. -/
theorem source_history_is_residue :
    SourceResidue translate .hotBigBangHistory := by
  rfl

/-- A representative source-side mechanism-condition residue witness. -/
theorem source_cp_condition_is_residue :
    SourceResidue translate .cAndCpViolation := by
  rfl

/-- A representative target-side methodological residue witness. -/
theorem target_jurisdiction_is_residue :
    TargetResidue translate .jurisdictionRecord := by
  intro a
  cases a <;> decide

```

```
/-- A second target-side methodological residue witness. -/
theorem target_residue_record_is_residue :
    TargetResidue translate .residueRecord := by
  intro a
  cases a <;> decide

/-- A registry uniquely generates `p` under a declared relation when every
    generator of `p` equals that registry. -/
def UniqueGenerator {Registry Problem : Type}
    (Generates : Registry -> Problem -> Prop)
    (g : Registry) (p : Problem) : Prop :=
  forall g', Generates g' p -> g' = g

/-- A problem is the unique problem admitted by a datum when every declared
    registry/problem pair generated from that datum yields that problem. -/
def UniqueProblemFrom {Registry Datum Problem : Type}
    (Generates : Registry -> Datum -> Problem -> Prop)
    (d : Datum) (p : Problem) : Prop :=
  forall g' p', Generates g' d p' -> p' = p

/-- Solving a problem does not certify uniqueness of its generator when a
    distinct generating registry is supplied as a counter-witness. -/
theorem solved_does_not_certify_unique_generator
    {Registry Problem : Type}
    (Solved : Problem -> Prop)
    (Generates : Registry -> Problem -> Prop)
    (g g' : Registry) (p : Problem)
    (hSolved : Solved p)
    (hGenerates : Generates g p)
    (hOtherGenerates : Generates g' p)
    (hDistinct : Not (g' = g)) :
    Solved p /\ Generates g p /\
      Not (UniqueGenerator Generates g p) := by
  refine And.intro hSolved (And.intro hGenerates ?_)
```

```
  intro hUnique
  exact hDistinct (hUnique g' hOtherGenerates)

/-- Solving one problem does not certify that it is the unique problem formed
    from the datum when a distinct generated problem is supplied as a witness. -/
theorem solved_does_not_certify_unique_problem
    {Registry Datum Problem : Type}
    (Solved : Problem -> Prop)
    (Generates : Registry -> Datum -> Problem -> Prop)
    (g g' : Registry) (d : Datum) (p p' : Problem)
    (hSolved : Solved p)
    (hGenerates : Generates g d p)
    (hOtherGenerated : Generates g' d p')
    (hDistinct : Not (p' = p)) :
    Solved p /\ Generates g d p /\
      Not (UniqueProblemFrom Generates d p) := by
  refine And.intro hSolved (And.intro hGenerates ?_)
  intro hUnique
  exact hDistinct (hUnique g' p' hOtherGenerated)

/-- Minimal compatibility at a registered anchor. -/
def CompatibleAt {Output : Type}
    (anchor outLeft outRight : Output) : Prop :=
  outLeft = anchor /\ outRight = anchor

/-- An explicit compatibility witness when both scoped outputs equal the same
    registered anchor. -/
theorem joint_closure_witness
    {Output : Type} (anchor outLeft outRight : Output)
    (hLeft : outLeft = anchor) (hRight : outRight = anchor) :
    CompatibleAt anchor outLeft outRight := by
  exact And.intro hLeft hRight

#check registered_anchor_invariant
```

```
#check source_history_is_residue
#check target_jurisdiction_is_residue
#check solved_does_not_certify_unique_generator
#check solved_does_not_certify_unique_problem
#check joint_closure_witness

#eval Lean.versionString

end LambdaCompanion
```

References

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Publication colophon

Release file	Function
Lambda_as_Complementary_Modeling_AUDITED_COMPANION_R2.pdf	Principal fixed-layout publication artifact
LambdaCompanion_Audit.lean	Exact executable formal source
BUILD_REPORT.txt	Human-readable build, verification, and scope record
CERTIFICATION_MANIFEST.json	Structured artifact, scope, source, and verification metadata
LEAN_WEB_VERIFICATION_EVIDENCE.json	Machine-readable public Lean verification evidence
SHA256SUMS.txt	Final post-freeze byte-identity receipt

> RELEASE BOUNDARY The release is ready for preservation and external review. Independent peer review and independent subject-matter validation remain future adjudications and are not implied by the word audited.